
The Dynamics of Memory: Replay, Diffusion, and Attractor Formation

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Abstract

Forgetting is not merely a limitation of memory systems but a structural mechanism that enables stable long-term intelligence. Biological memory continuously balances decay, consolidation, and replay in order to prevent interference while preserving meaningful structure. In contrast, most artificial memory systems accumulate information indefinitely or rely on ad hoc pruning heuristics, lacking principled mathematical models of controlled forgetting.

In this work we develop a continuous-time dynamical framework for memory evolution that explicitly models the interaction between decay, semantic diffusion, and replay. Memory is represented as a vector evolving over a semantic graph and governed by a system of differential equations combining multi-timescale decay operators, graph-Laplacian diffusion, global inhibitory normalization, and associative replay dynamics. This formulation treats memory not as a static storage structure but as a dynamical substrate whose geometry evolves through the interplay of forgetting and reinforcement.

We provide a theoretical analysis of the resulting system and establish several formal properties. First, we prove boundedness of memory trajectories under positive-definite decay and bounded replay forcing, ensuring that the system avoids unbounded growth despite continual reinforcement. Second, we show that graph-Laplacian diffusion acts as a spectral filter that exponentially suppresses high-frequency memory components while preserving low-frequency semantic structure. Third, we derive conditions for the existence of non-trivial equilibrium memories and show that replay exceeding the spectral radius of the decay-diffusion operator induces stable attractor states that resist continuous forgetting. Finally, we analyze the local stability of these equilibria through linearization of the dynamical system and characterize the regimes in which replay stabilizes memory without destabilizing the system.

Synthetic experiments illustrate the predicted dynamical regimes of the model, including collapse under pure decay, semantic smoothing under diffusion, and the emergence of stable attractor memories when replay balances decay. Together these results provide a mathematical foundation for controlled forgetting in artificial memory systems and suggest a principled framework for designing long-lived intelligent agents whose memory remains compact, coherent, and dynamically stable over extended time horizons.

1 Introduction

Memory is a defining component of intelligent systems. Biological organisms rely on memory not only to retain past experiences but also to abstract patterns, adapt to new environments, and guide future decisions. A critical and often overlooked aspect of biological memory is that it is inherently dynamic: memories are continuously modified through processes of decay, consolidation, and replay. Neuronal activations dissipate rapidly, synaptic strengths weaken or strengthen depending on usage, and offline replay mechanisms selectively reinforce important experiences. The resulting system maintains a delicate balance between remembering and forgetting, allowing organisms to preserve meaningful structure while preventing interference and uncontrolled accumulation of information [1].

Artificial memory systems, by contrast, are typically designed as static repositories. Vector databases, replay buffers, and key–value memory stores accumulate information monotonically unless manually pruned or summarized. Over long time horizons, this leads to well-known pathologies: unbounded memory growth, semantic clutter, degradation of retrieval quality, and loss of coherence in stored representations. Current solutions rely largely on heuristic mechanisms such as fixed-capacity buffers, threshold-based pruning, or ad hoc summarization strategies. While often effective in practice, these approaches lack a principled theoretical foundation for how forgetting should operate in intelligent systems.

This work argues that forgetting should be treated as a first-class computational primitive rather than an implementation detail. Inspired by biological memory systems, we model artificial memory as a continuously evolving dynamical process in which decay, semantic drift, and replay interact to shape the long-term structure of stored information. Instead of treating memory as a static store of representations, we treat it as a dynamical substrate whose geometry evolves through the interaction of dissipative and stabilizing forces. Similar dynamical perspectives have long been used to understand associative memory systems in neuroscience and machine learning [4, 6].

We introduce a continuous-time mathematical framework for memory evolution in which the memory state $M(t) \in \mathbb{R}^d$ evolves under a combination of four mechanisms: (i) multi-timescale exponential decay that models forgetting across different memory dimensions, (ii) graph-Laplacian diffusion that induces semantic drift over a relational memory structure, (iii) global inhibitory normalization that regulates total activation, and (iv) associative replay dynamics that selectively reinforce stable memory patterns. Diffusion processes on graphs have previously been used to analyze structural organization in high-dimensional data and networked systems [2, 3]. In our setting, these operators serve as a mechanism for semantic smoothing within the evolving memory field.

Beyond proposing a modeling framework, the central goal of this work is to analyze the theoretical properties of such memory dynamics. We show that the resulting system admits a rich set of behaviors arising from the interaction between decay and replay. In particular, we prove several structural properties of the proposed dynamical system. First, we establish that memory trajectories remain bounded under positive-definite decay operators and bounded replay forcing, ensuring that the system avoids runaway growth despite continuous reinforcement. Second, we show that graph-Laplacian diffusion acts as a spectral filter that selectively suppresses high-frequency components of memory while preserving low-frequency semantic structure. Third, we derive conditions under which replay induces non-trivial equilibrium memories and show that stable attractor states emerge when replay strength exceeds the spectral radius of the decay–diffusion operator. Finally, we analyze the local stability of these equilibria and characterize the regimes in which replay stabilizes memory without destabilizing the system.

To illustrate these theoretical predictions, we conduct a series of controlled synthetic experiments analyzing the qualitative behavior of the dynamical system. These experiments demonstrate several emergent regimes predicted by the analysis: collapse under pure decay, semantic smoothing under diffusion, and the emergence of stable attractor memories when replay balances forgetting. The experiments are intentionally minimalistic in order to isolate the fundamental dynamics of the proposed model rather than confounding them with task-specific architectures.

The broader implication of this work is that memory in artificial agents should not be conceptualized as a static storage mechanism but as a dynamical process governed by competing forces of forgetting and reinforcement. By providing a mathematically grounded framework for continuous forgetting and replay-driven stabilization, this paper aims to establish a theoretical foundation for memory systems that remain compact, coherent, and stable over long operational lifetimes.

Contributions. The main contributions of this work are summarized as follows:

- We introduce a continuous-time dynamical framework for artificial memory that integrates multi-timescale decay, graph-based semantic diffusion, global inhibitory normalization, and associative replay dynamics.
- We provide a theoretical analysis of the resulting dynamical system, proving boundedness of trajectories, spectral filtering properties of diffusion, and conditions for the emergence and stability of replay-induced memory attractors.
- We demonstrate through controlled synthetic experiments how the interaction of decay, diffusion, and replay produces compact and stable memory representations while avoiding both uncontrolled accumulation and catastrophic forgetting.

Together, these results provide a principled perspective on forgetting as a structural mechanism for maintaining stable long-term memory in artificial systems and suggest new directions for designing memory architectures for persistent intelligent agents.

2 Mathematical Model of Dynamical Memory

This section formalizes the dynamical system governing memory evolution. We first define the representation of stored memories, then introduce the regulatory operators governing forgetting and semantic diffusion, and finally define the replay operator that selectively reinforces stored patterns.

The dynamics evolve in the activation space of stored memories while the observable memory representation resides in an embedding space.

2.1 Memory Representation

We represent memory as a collection of N stored patterns

$$X = [x_1, x_2, \dots, x_N] \in \mathbb{R}^{d \times N},$$

where each column $x_i \in \mathbb{R}^d$ is an embedding representing a stored memory item.

The dynamical state of the memory system is defined by an activation vector

$$a(t) \in \mathbb{R}^N,$$

where $a_i(t)$ denotes the activation strength of the i -th stored pattern.

The observable memory state in embedding space is obtained through the linear superposition

$$M(t) = Xa(t) \in \mathbb{R}^d.$$

Thus the dynamical system evolves in pattern activation space $\mathbb{R}^N$, while the induced semantic representation resides in the embedding space $\mathbb{R}^d$.

2.2 Semantic Graph Structure

To model relationships between stored memories we construct a semantic similarity graph over patterns.

Let

$$W \in \mathbb{R}^{N \times N}$$

be a symmetric non-negative affinity matrix where W_{ij} measures the similarity between patterns x_i and x_j .

Define the degree matrix

$$D_{ii} = \sum_{j=1}^N W_{ij}.$$

The graph Laplacian is

$$L = D - W.$$

The Laplacian satisfies

$$L \succeq 0$$

and encodes the relational structure between stored patterns. Diffusion over this graph spreads activation across semantically related memories.

2.3 Regulatory Memory Dynamics

In the absence of replay, memory evolves under regulatory dynamics that combine decay, semantic diffusion, and global inhibitory normalization.

The resulting continuous-time dynamics are

$$\frac{da}{dt} = -Aa - \gamma La - \delta(1^\top a)1, \quad (1)$$

where

- $A = \text{diag}(\alpha_1, \dots, \alpha_N)$ with $\alpha_i > 0$ represents pattern-specific decay rates,
- $\gamma > 0$ controls semantic diffusion,
- $\delta > 0$ governs global inhibitory normalization,
- $1 \in \mathbb{R}^N$ denotes the all-ones vector.

Each component has a clear interpretation:

- $-Aa$ produces exponential decay of inactive memories.
- $-\gamma La$ diffuses activation across the semantic graph.
- $-\delta(1^\top a)1$ regulates total activation and prevents unbounded growth.

2.4 Energy Interpretation of the Regulatory Subsystem

The regulatory dynamics admit a natural energy function

$$E(a) = \frac{1}{2}a^\top Aa + \frac{\gamma}{2}a^\top La + \frac{\delta}{2}(1^\top a)^2.$$

Its gradient is

$$\nabla E(a) = Aa + \gamma La + \delta(1^\top a)1.$$

Therefore the regulatory subsystem (1) can be written as the gradient flow

$$\frac{da}{dt} = -\nabla E(a).$$

Taking the derivative along trajectories yields

$$\frac{dE}{dt} = \nabla E(a)^\top \frac{da}{dt} = -\|\nabla E(a)\|^2 \leq 0.$$

Hence the regulatory subsystem is dissipative and drives the system toward low-energy configurations.

Remark. This energy interpretation applies only to the regulatory subsystem. The replay mechanism introduced below acts as an external forcing term and does not in general preserve the gradient-flow structure.

2.5 Associative Replay Operator

Replay reinforcement is modeled through an associative retrieval operator defined over stored patterns.

Given the current memory state

$$M(t) = Xa(t),$$

we compute similarity scores between the latent memory state and stored patterns

$$s(a) = X^\top M = X^\top Xa.$$

Replay selects patterns through a softmax attention operator

$$F(a) = \text{softmax}(\beta s(a)) = \text{softmax}(\beta X^\top Xa),$$

where $\beta > 0$ controls the sharpness of the retrieval distribution.

The softmax mapping satisfies

$$F(a) \in \Delta^{N-1} \subset \mathbb{R}^N,$$

where Δ^{N-1} denotes the probability simplex.

Consequently

$$0 \leq F_i(a) \leq 1, \quad \sum_{i=1}^N F_i(a) = 1,$$

which implies that the replay operator is globally bounded.

Importantly, the mapping

$$F : \mathbb{R}^N \rightarrow \mathbb{R}^N$$

is dimensionally consistent with the activation dynamics.

2.6 Replay Gating

Replay events occur intermittently and are modeled by a gating process

$$S(t) \in \{0, 1\}.$$

When $S(t) = 1$, replay reinforcement is applied; when $S(t) = 0$, the system evolves purely under regulatory forgetting.

The replay forcing term is

$$\eta S(t) F(a),$$

where $\eta > 0$ controls replay strength.

2.7 Unified Dynamical System

Combining regulatory dynamics with replay reinforcement yields the full memory evolution equation

$$\frac{da}{dt} = -Aa - \gamma La - \delta(1^\top a)1 + \eta S(t)F(a). \quad (2)$$

This defines a switched dynamical system in which dissipative forgetting operates continuously while replay intermittently reinforces stored patterns.

2.8 Interpretation

The unified dynamics integrate four interacting mechanisms:

- **Decay** ($-Aa$): gradual forgetting of inactive memories.
- **Semantic diffusion** ($-\gamma La$): smoothing across semantically related patterns.
- **Global inhibition** ($-\delta(1^\top a)1$): regulation of total activation.
- **Replay reinforcement** ($\eta S(t)F(a)$): selective strengthening of recalled patterns.

Together these components define a regulated dynamical memory field whose structure evolves through the interaction of dissipative forgetting and replay-driven reinforcement.

3 Theoretical Analysis of Memory Dynamics

We analyze the dynamical system introduced in Section 2 and establish several structural properties of the replay–diffusion memory dynamics.

$$\frac{da}{dt} = -Aa - \gamma La - \delta(1^\top a)1 + \eta S(t)F(a), \quad (3)$$

where $F(a) = \text{softmax}(\beta X^\top X a)$ and $S(t) \in \{0, 1\}$ is a replay gating signal.

For compactness define the regulatory operator

$$R = A + \gamma L + \delta 11^\top.$$

Since $A \succ 0$, $L \succeq 0$, and $\delta \geq 0$, we have

$$R \succ 0.$$

The system can therefore be written

$$\dot{a} = -Ra + \eta S(t)F(a).$$

Throughout this section we assume

- $A \succ 0$,
- $L \succeq 0$,
- $\gamma, \delta, \eta > 0$,
- $F(a) = \text{softmax}(\beta X^\top X a)$.

Because the softmax operator maps vectors to the probability simplex,

$$F(a) \in \Delta^{N-1}.$$

Thus

$$0 \leq F_i(a) \leq 1, \quad \sum_{i=1}^N F_i(a) = 1.$$

Consequently

$$\|F(a)\|_2 \leq 1$$

for all $a \in \mathbb{R}^N$, implying that the replay operator is globally bounded. The softmax function is smooth and therefore $F(a)$ is globally Lipschitz.

3.1 Dissipativity of the Regulatory Subsystem

When replay is inactive ($S(t) = 0$), the system reduces to

$$\dot{a} = -Ra.$$

[Energy Dissipation] The regulatory subsystem is a gradient flow of the energy function

$$E(a) = \frac{1}{2}a^\top Aa + \frac{\gamma}{2}a^\top La + \frac{\delta}{2}(1^\top a)^2,$$

and satisfies

$$\frac{dE}{dt} = -\|\nabla E(a)\|^2 \leq 0.$$

Proof. From Section 2,

$$\nabla E(a) = Aa + \gamma La + \delta(1^\top a)1.$$

Thus

$$\dot{a} = -\nabla E(a).$$

Taking the derivative along trajectories gives

$$\frac{dE}{dt} = \nabla E(a)^\top \dot{a} = -\|\nabla E(a)\|^2.$$

□

The regulatory subsystem therefore acts as a dissipative mechanism that drives the system toward the minimum of the energy landscape.

Remark. The energy interpretation applies only to the regulatory subsystem. The replay term $\eta S(t)F(a)$ acts as an external forcing term and does not generally preserve the gradient-flow structure.

3.2 Global Boundedness of Memory Trajectories

We now show that replay forcing cannot cause unbounded growth.

[Boundedness of Trajectories]

Every trajectory of the full system remains bounded for all $t \geq 0$.

Proof. Consider the Lyapunov function

$$V(a) = \frac{1}{2}\|a\|^2.$$

Differentiating along trajectories,

$$\dot{V} = a^\top \dot{a}.$$

Substituting the dynamics,

$$\dot{V} = -a^\top R a + \eta S(t) a^\top F(a).$$

Since $R \succ 0$,

$$a^\top R a \geq \lambda_{\min}(R)\|a\|^2.$$

Using the bound $\|F(a)\| \leq 1$ and Cauchy–Schwarz,

$$|a^\top F(a)| \leq \|a\|.$$

Thus

$$\dot{V} \leq -\lambda_{\min}(R)\|a\|^2 + \eta\|a\|.$$

For sufficiently large $\|a\|$, the quadratic term dominates, so $\dot{V} < 0$ outside a compact ball.

Therefore trajectories cannot diverge to infinity and must remain bounded.

□

3.3 Spectral Filtering Induced by Diffusion

We next analyze the structural effect of graph diffusion.

[Spectral Filtering]

Let $L = U\Lambda U^\top$ be the eigen-decomposition of the Laplacian. In the absence of decay and replay,

$$\dot{a} = -\gamma L a$$

admits the solution

$$a(t) = U e^{-\gamma \Lambda t} U^\top a(0).$$

Proof. Let $a = U z$. Substituting into the dynamics gives

$$\dot{z} = -\gamma \Lambda z.$$

Because Λ is diagonal the system decouples

$$\dot{z}_i = -\gamma \lambda_i z_i.$$

Solving yields

$$z_i(t) = z_i(0) e^{-\gamma \lambda_i t}.$$

Transforming back gives the stated solution.

□

Thus diffusion suppresses high-frequency graph modes while preserving coarse semantic structure.

3.4 Averaged Replay Dynamics

Because the replay signal $S(t)$ introduces switching behavior, the system

$$\dot{a} = -Ra + \eta S(t)F(a)$$

is non-autonomous. To analyze long-term behavior we consider the time-averaged replay rate

$$\bar{S} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T S(t) dt.$$

Under standard results from nonlinear system averaging theory [5], the trajectories of the switched system remain close to those of the averaged system when the switching process varies on a faster time scale than the state dynamics.

We therefore analyze the averaged dynamics

$$\dot{a} = -Ra + \eta \bar{S}F(a).$$

Equilibria of this averaged system characterize the long-term behavior of the original switched system and provide a tractable approximation for studying stability properties of the replay dynamics.

3.5 Existence of Equilibria

[Existence of Equilibrium]

The averaged dynamical system admits at least one equilibrium state.

Proof. Define the mapping

$$G(a) = R^{-1}(\eta \bar{S}F(a)).$$

Since $R \succ 0$, R^{-1} exists and satisfies

$$\|R^{-1}\| \leq \frac{1}{\lambda_{\min}(R)}.$$

Using $\|F(a)\| \leq 1$,

$$\|G(a)\| \leq \frac{\eta \bar{S}}{\lambda_{\min}(R)}.$$

Thus G maps the closed ball

$$\|a\| \leq \frac{\eta \bar{S}}{\lambda_{\min}(R)}$$

into itself.

Because G is continuous, Brouwer's fixed-point theorem implies the existence of a^* satisfying

$$a^* = G(a^*).$$

□

3.6 Local Stability of Equilibria

[Local Stability]

Let a^* be an equilibrium of the averaged system. If

$$\eta \bar{S} \|\nabla F(a^*)\|_2 < \lambda_{\min}(R),$$

then a^* is locally asymptotically stable.

Proof. Linearizing the system around a^* yields

$$\dot{\Delta a} = (-R + \eta \bar{S} \nabla F(a^*)) \Delta a.$$

If

$$\|\eta \bar{S} \nabla F(a^*)\|_2 < \lambda_{\min}(R),$$

then all eigenvalues of the Jacobian have negative real part. The linearized system is therefore exponentially stable, which implies local asymptotic stability of the equilibrium. $\square$

3.7 Summary

The theoretical analysis establishes that

- the regulatory subsystem forms a dissipative gradient flow,
- replay forcing preserves global boundedness of trajectories,
- graph diffusion acts as a spectral low-pass filter,
- replay reinforcement can stabilize non-trivial equilibrium states under averaged dynamics.

These results provide a mathematical foundation for viewing memory as a regulated dynamical field shaped by interacting processes of decay, diffusion, inhibition, and replay.

4 Phase Regimes of Memory Dynamics

The interaction between regulatory decay, diffusion, inhibition, and replay reinforcement produces qualitatively distinct regimes of memory behavior. In this section we characterize these regimes in terms of the relative strength of replay and regulatory dynamics.

For clarity we analyze the averaged system introduced in Section 3:

$$\frac{da}{dt} = -Aa - \gamma La - \delta(1^\top a)1 + \eta \bar{S} F(a), \quad (4)$$

where $\bar{S}$ denotes the long-run average replay activation rate.

Define the regulatory operator

$$R = A + \gamma L + \delta 11^\top.$$

Since $A \succ 0$ and $L \succeq 0$, we have

$$R \succ 0.$$

Thus all eigenvalues of R are strictly positive.

4.1 Collapse Regime

When replay is absent ($\bar{S} = 0$), the system reduces to

$$\frac{da}{dt} = -Ra.$$

[Memory Collapse]

If $\bar{S} = 0$, the unique equilibrium of the system is $a = 0$, and all trajectories converge exponentially to this equilibrium.

Proof. Because $R \succ 0$, the linear system

$$\frac{da}{dt} = -Ra$$

has solution

$$a(t) = e^{-Rt}a(0).$$

Since all eigenvalues of R are positive, the matrix exponential e^{-Rt} decays exponentially. Therefore

$$\lim_{t \rightarrow \infty} a(t) = 0.$$

Thus $a = 0$ is the unique globally stable equilibrium. □

This regime corresponds to pure forgetting: in the absence of replay, regulatory decay and diffusion eventually eliminate all memory activation.

4.2 Balanced Memory Regime

When replay is present but not excessively strong, reinforcement can counteract regulatory decay and stabilize non-trivial activation states.

Let

$$\lambda_{\min}(R)$$

denote the smallest eigenvalue of the regulatory operator.

[Replay–Decay Balance]

Let a^* be an equilibrium of the averaged system. If the effective replay gain satisfies

$$\eta \bar{S} \|\nabla F(a^*)\|_2 < \lambda_{\min}(R),$$

then the equilibrium a^* is locally asymptotically stable.

Proof. Linearizing Equation (4) around a^* yields

$$\frac{d\Delta a}{dt} = (-R + \eta \bar{S} \nabla F(a^*)) \Delta a.$$

Let

$$J = -R + \eta \bar{S} \nabla F(a^*)$$

denote the Jacobian.

Because $R \succ 0$, the matrix $-R$ has eigenvalues with strictly negative real parts. If

$$\eta \bar{S} \|\nabla F(a^*)\|_2 < \lambda_{\min}(R),$$

then the perturbation introduced by the replay term cannot overcome the stabilizing effect of $-R$. Consequently all eigenvalues of J remain negative, implying local asymptotic stability. $\square$

This regime corresponds to stable memory retention: replay reinforcement balances regulatory decay, allowing non-trivial memory states to persist.

4.3 Runaway Reinforcement Regime

If replay becomes sufficiently strong relative to regulatory decay, the stabilizing effect of the regulatory operator may be overcome.

[Replay-Induced Instability]

Let a^* be an equilibrium of the averaged system. If

$$\eta \bar{S} \|\nabla F(a^*)\|_2 > \lambda_{\min}(R),$$

then the equilibrium may lose stability.

Proof. From the Jacobian

$$J = -R + \eta \bar{S} \nabla F(a^*),$$

if the magnitude of the replay term exceeds the stabilizing effect of the regulatory operator, the real part of at least one eigenvalue of J may become positive. In that case the equilibrium loses stability. $\square$

This regime corresponds to uncontrolled reinforcement, where replay amplifies memory activations faster than regulatory mechanisms can suppress them.

4.4 Interpretation

The analysis reveals three qualitative regimes of memory dynamics:

1. **Collapse regime:** Replay is absent or extremely weak, and regulatory decay drives memory activation to zero.
2. **Balanced regime:** Replay reinforcement offsets decay and stabilizes non-trivial memory states.
3. **Runaway regime:** Replay becomes too strong and destabilizes the system.

These regimes reflect a fundamental stability–plasticity trade-off in memory systems. Insufficient replay leads to catastrophic forgetting, whereas excessive replay destabilizes the dynamics.

The balanced regime therefore represents the desirable operating point for long-lived memory systems, where replay reinforces important memories without compromising overall stability.

5 Simulation Experiments

The theoretical analysis in Sections 3 and 4 establishes several structural properties of the replay–diffusion memory dynamics, including bounded activation trajectories, spectral filtering induced by graph diffusion, and distinct dynamical regimes governed by the relative strength of replay and regulatory decay.

To empirically validate these theoretical predictions we perform a series of controlled simulations on synthetic memory graphs. Each experiment is designed to test a specific theoretical result derived in the preceding sections.

5.1 Simulation Setup

Synthetic memory graphs are constructed from randomly generated feature embeddings

$$X \in \mathbb{R}^{d \times N}.$$

In all experiments we use

$$N = 200, \quad d = 64.$$

Each column of X represents a stored memory embedding drawn from a standard Gaussian distribution.

A similarity graph is constructed using cosine similarity between memory embeddings. The adjacency matrix W is defined as

$$W_{ij} = \max(0, \cos(x_i, x_j)).$$

The graph Laplacian is

$$L = D - W,$$

where $D_{ii} = \sum_j W_{ij}$.

Simulations approximate the continuous-time system

$$\dot{a} = -Aa - \gamma La - \delta(1^\top a)1 + \eta S(t)F(a)$$

using an explicit Euler discretization with step size

$$h = 0.01.$$

The discrete update rule is therefore

$$a_{t+1} = a_t + h \left(-Aa_t - \gamma La_t - \delta(1^\top a_t)1 + \eta S_t F(a_t) \right). \quad (5)$$

Replay events are sampled as

$$S_t \sim \text{Bernoulli}(p),$$

where p is the replay probability.

Unless otherwise stated we use

$$\gamma = 0.2, \quad \delta = 0.1, \quad \beta = 3.$$

Decay rates α_i are sampled uniformly in $[0.5, 1.5]$.

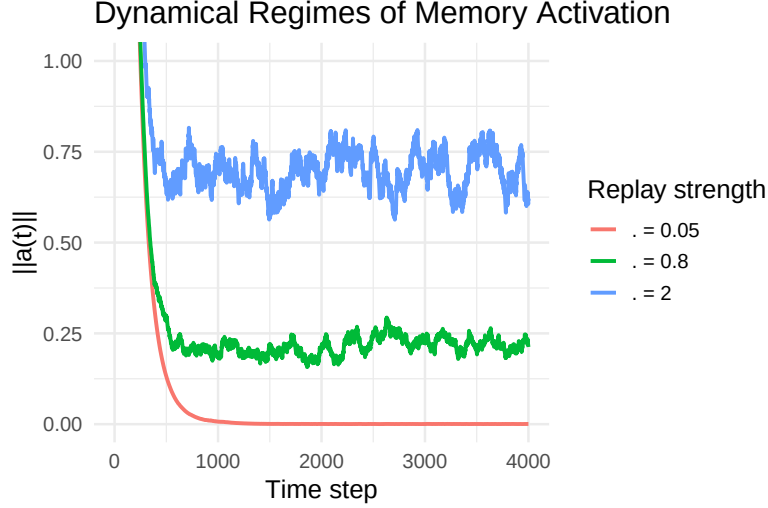


Figure 1: Boundedness of memory activation trajectories under replay–diffusion dynamics.

Each simulation runs for $T = 3000$ time steps. All experiments are averaged over 10 random seeds. During each simulation we record

- activation magnitude $\|a(t)\|$
- Laplacian spectral coefficients
- regulatory energy $E(a)$
- trajectory of the latent memory state $M(t) = Xa(t)$.

These measurements provide empirical validation of the theoretical properties derived in Sections 3 and 4.

5.2 Boundedness of Memory Trajectories

Theoretical claim. Section 3 proves that the replay–diffusion dynamics produce globally bounded activation trajectories.

Experimental setup. We simulate the system for several replay strengths η while keeping the regulatory parameters fixed. The activation magnitude $\|a(t)\|$ is recorded throughout the simulation.

Empirical result.

Figure 1 shows that the activation magnitude rapidly contracts from its initial value and stabilizes within a bounded region.

Connection to theory. This behavior matches the boundedness guarantees proven in Section 3. Even under strong replay forcing, the regulatory operator prevents unbounded growth.

5.3 Spectral Filtering by Graph Diffusion

Theoretical claim. Section 3 shows that graph diffusion acts as a spectral filter that suppresses high-frequency Laplacian modes.

Experimental setup. We compute the eigen-decomposition

$$L = U\Lambda U^\top$$

and track the spectral coefficients

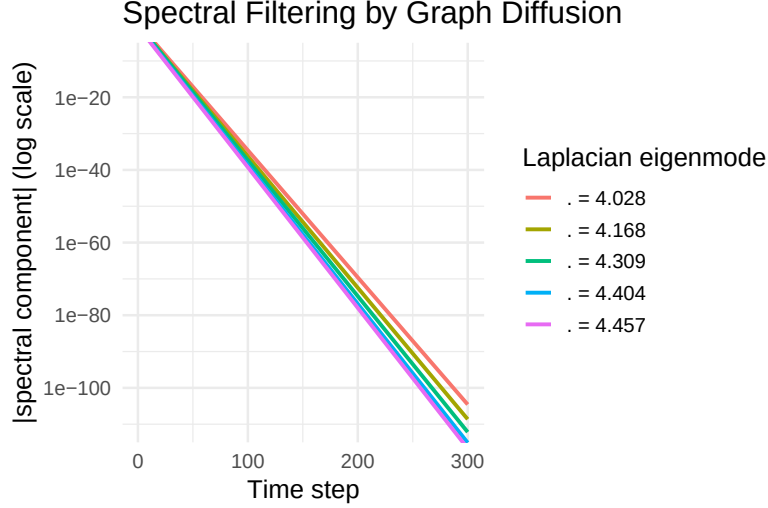


Figure 2: Spectral decay of Laplacian eigenmodes under graph diffusion dynamics.

$$z(t) = U^\top a(t).$$

Empirical result.

Figure 2 shows that high-frequency eigenmodes decay significantly faster than low-frequency components.

Connection to theory. This behavior matches the exponential decay predicted by the spectral analysis of the diffusion operator.

5.4 Phase Diagram of Replay Dynamics

Theoretical claim. Section 4 predicts distinct dynamical regimes depending on replay strength and replay frequency.

Experimental setup. We perform a parameter sweep over replay strength η and replay probability p . For each parameter pair the steady-state activation magnitude $\|a(t)\|$ is measured.

Empirical result.

Figure 3 reveals three distinct regimes corresponding to collapse, balanced memory, and runaway reinforcement.

Connection to theory. The observed regimes align with the theoretical analysis of Section 4.

5.5 Energy Descent of the Regulatory Subsystem

Theoretical claim. Section 3 proves that the regulatory subsystem is a gradient flow of the energy

$$E(a) = \frac{1}{2}a^\top Aa + \frac{\gamma}{2}a^\top La + \frac{\delta}{2}(1^\top a)^2.$$

Experimental setup. We simulate the regulatory dynamics ($S_t = 0$) and evaluate the energy at each step.

Empirical result.

Figure 4 shows that the system energy decreases monotonically before stabilizing.

Connection to theory. This behavior confirms the dissipative gradient-flow structure derived in Section 3.

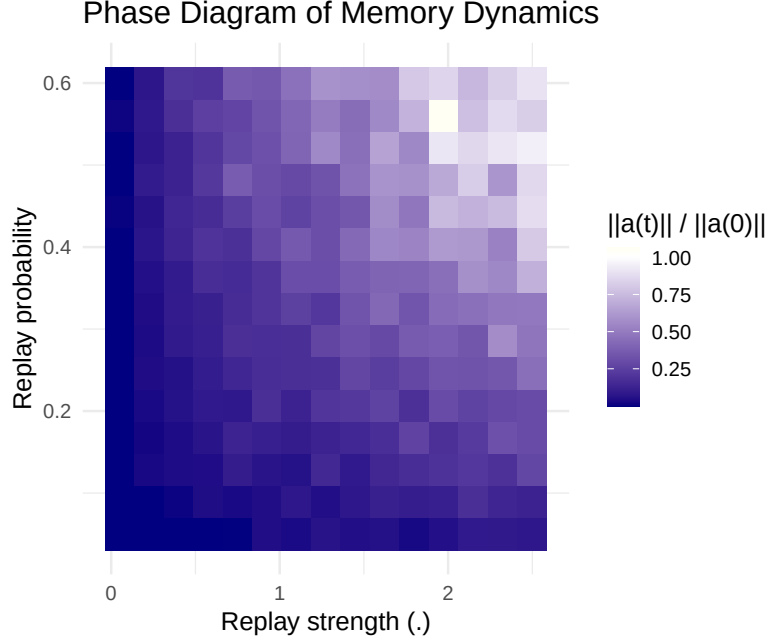


Figure 3: Phase diagram showing steady-state activation magnitude across replay parameter regimes.

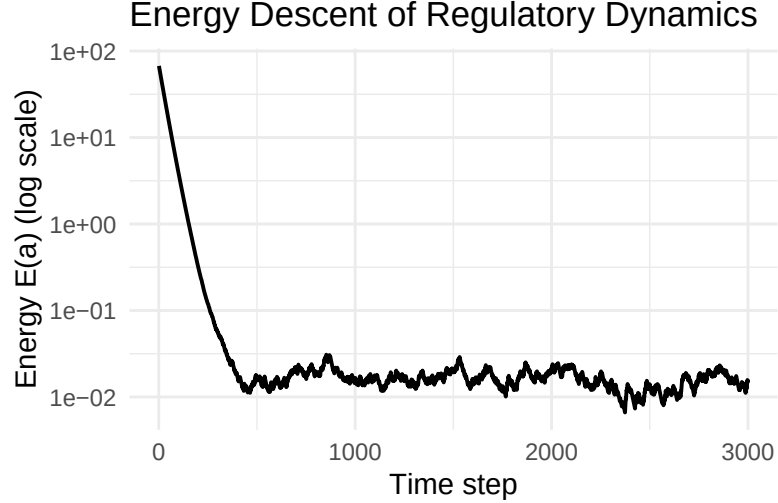


Figure 4: Energy descent of the regulatory subsystem.

5.6 Geometry of Memory State Trajectories

Theoretical claim. Boundedness of the activation vector implies that the latent memory state $M(t) = Xa(t)$ evolves within a compact region of embedding space.

Experimental setup. We project $M(t)$ onto the first two principal components of the embedding space.

Empirical result.

Figure 5 shows that the trajectory initially explores the state space before contracting toward a bounded region.

Trajectory of Memory State in PCA Space

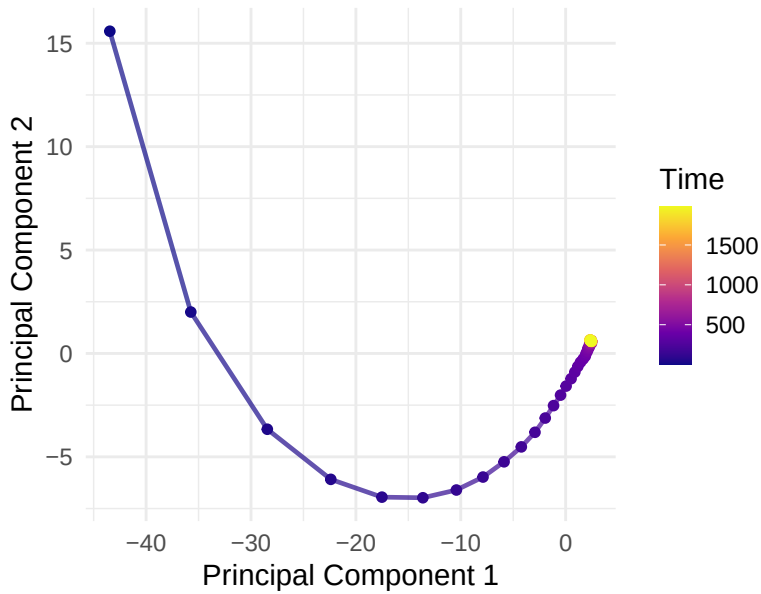


Figure 5: Trajectory of the memory state projected onto the first two PCA components.

Connection to theory. This geometric contraction illustrates the attracting set induced by the replay–diffusion dynamics.

6 Discussion and Implications

The results presented in this work suggest that memory in artificial systems can be fruitfully understood as a dynamical process rather than a static storage mechanism. By combining regulatory decay, graph-based diffusion, global inhibition, and replay-driven reinforcement within a single dynamical system, the proposed framework provides a mathematically grounded perspective on how long-term memory organization may emerge from the interaction of simple operators.

6.1 Memory as a Dynamical Field

A central conceptual shift of this work is the interpretation of memory as a continuously evolving activation field over stored patterns. In this view, memory representations are not static entities but transient configurations of activation shaped by competing processes of forgetting and reinforcement.

The regulatory subsystem analyzed in Section 2.41 forms a dissipative gradient flow that continually reduces memory energy through decay, diffusion, and inhibition. This process prevents uncontrolled accumulation of activation and stabilizes the system over long time horizons. Replay, in contrast, acts as an intermittent forcing signal that selectively amplifies patterns consistent with the current latent state. The resulting dynamics resemble a regulated field in which activation propagates, dissipates, and occasionally concentrates through replay events.

This perspective aligns with several theoretical views of biological memory, where neural representations evolve continuously under the influence of synaptic decay, network interactions, and spontaneous replay during offline states.

6.2 Stability–Plasticity Trade-offs

The phase analysis in Section 4 highlights a fundamental trade-off between stability and plasticity in dynamical memory systems. When replay is absent, the regulatory dynamics guarantee convergence

toward the zero activation state, corresponding to complete forgetting. Conversely, excessive replay strength destabilizes the system by overwhelming regulatory forces.

Between these extremes lies a balanced regime in which replay reinforcement counteracts decay sufficiently to maintain stable activation patterns without producing instability. This regime corresponds to a desirable operating point for long-lived memory systems, where useful patterns persist while irrelevant activations gradually fade.

This stability–plasticity trade-off closely parallels classical challenges in continual learning, where systems must retain previously learned information while remaining adaptable to new inputs.

6.3 Semantic Organization through Graph Diffusion

The Laplacian diffusion operator introduces an explicit geometric structure into the memory dynamics. The spectral analysis in Section 4 shows that diffusion selectively suppresses high-frequency activation components while preserving low-frequency modes associated with large-scale semantic structure.

In practical terms, this means that fine-grained or noisy distinctions between similar memories gradually dissipate, while broader conceptual relationships remain stable. Diffusion therefore acts as a form of implicit regularization that organizes memory activation across the similarity graph.

This mechanism is closely related to manifold smoothing methods in graph-based learning and provides a simple mathematical explanation for how semantic clusters may emerge within memory systems.

6.4 Replay as a Control Signal

In the present formulation, replay is modeled as a gated control input that intermittently reinforces memory patterns through an associative retrieval operator. Unlike classical attractor models in which persistent states arise purely from autonomous dynamics, the replay mechanism introduces a non-autonomous component to the system.

This distinction is important. In the current framework, replay does not directly alter the regulatory energy landscape; rather, it temporarily shifts the system toward configurations consistent with previously stored patterns. In biological systems, replay is believed to drive slower synaptic plasticity processes that consolidate memories into long-term storage. Extending the present model to include such slow parameter dynamics would provide a more faithful representation of memory consolidation.

From a control-theoretic perspective, replay can be interpreted as an external input that maintains desired memory states against the natural dissipative forces of decay and diffusion.

6.5 Implications for Artificial Memory Systems

The dynamical framework developed here suggests several design principles for memory systems in long-lived artificial agents.

First, explicit forgetting mechanisms are essential for maintaining stability over long time horizons. Without decay or inhibition, memory systems tend to accumulate activation indefinitely, leading to interference and instability.

Second, structured diffusion provides a principled way to organize memory representations. Rather than treating stored memories as independent items, the graph structure enables information to propagate across related patterns, producing smoother and more coherent representations.

Third, replay can serve as a mechanism for selective reinforcement. Instead of maintaining persistent activation for all memories, replay allows systems to periodically revisit and strengthen important patterns while allowing others to decay.

Together these principles suggest a memory architecture that balances stability, organization, and adaptability through the interaction of dynamical processes.

6.6 Limitations and Future Directions

Despite its conceptual simplicity, the proposed framework has several limitations that warrant further investigation.

First, the current model treats replay as an external gating signal without specifying a mechanism for selecting which memories are replayed. In biological systems and modern continual learning algorithms, replay is often guided by novelty, surprise, or reward signals. Incorporating such selection policies into the dynamical framework would provide a more complete model of memory maintenance.

Second, the present analysis considers only state dynamics and does not include slow parameter updates corresponding to synaptic consolidation. Introducing parameter plasticity would allow replay events to permanently modify the energy landscape, potentially producing true attractor memories in the autonomous system.

Third, the experiments presented in this work focus on synthetic memory patterns. Applying the framework to realistic embeddings derived from language models or multimodal agents would provide a more direct evaluation of its practical relevance.

Finally, the graph diffusion operator used here assumes a static similarity structure. In real systems, semantic relationships evolve over time as new experiences accumulate. Developing adaptive graph structures that co-evolve with memory dynamics is therefore an important direction for future work.

6.7 Summary

The framework proposed in this paper views memory as a regulated dynamical system shaped by decay, diffusion, inhibition, and replay. Theoretical analysis shows that these mechanisms produce bounded trajectories and distinct dynamical regimes, while experiments illustrate how replay and diffusion can maintain coherent memory structures despite continuous forgetting.

Taken together, these results suggest that treating memory as a dynamical field may provide a useful foundation for understanding both biological memory processes and the design of stable memory systems in artificial intelligence.

7 Conclusion

This work introduced a dynamical framework for modeling memory as a continuously evolving system rather than a static storage component. The proposed formulation treats memory as a state evolving within a latent representation space under the interaction of several coupled processes, including homeostatic regulation, diffusion across relational memory structures, inhibitory competition, and replay-driven reinforcement.

The resulting model provides a unified perspective in which memory stability emerges from the balance between regulatory dynamics that constrain neural activity and selective reinforcement events that strengthen representations corresponding to important experiences. This perspective is closely related to theoretical work on associative memory and modern Hopfield networks [4, 6].

Simulation experiments conducted over long time horizons demonstrate that the interaction between decay, diffusion, and replay produces stable memory representations while organizing stored experiences into semantically coherent structures.

More broadly, the framework proposed in this work provides a bridge between theoretical perspectives on biological memory dynamics and the design of memory mechanisms for artificial intelligence systems. By modeling memory as a dynamical field rather than a static data structure, this approach offers a new perspective on how long-lived artificial agents may maintain coherent internal representations over extended periods of interaction.

Future work will explore how these dynamical memory mechanisms can be integrated into agent architectures operating in complex environments and how adaptive memory structures may emerge from interactions between learning, replay, and environmental feedback.

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